Aidar Begotayev

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EDUCATION

BSc Computer Science with Artificial Intelligence, Coventry University Kazakhstan, Astana; 2025 – present GPA 3.75/4

Nazarbayev Intellectual School of Astana, IB Diploma (Math AA HL); 2019 – 2025 IB 39/45, GPA 4.85

RESEARCH EXPERIENCE

Supervised Research, Coventry University Kazakhstan (supervisor: Sanzhar Kusdavletov); 2026 – present

- Investigating reinforcement learning combined with Hamilton–Jacobi–Bellman value functions for safety-critical control (UAV interception); proposed a decomposition V = V_HJB + learned residual, to be tested in simulation

Independent Researcher, Lomonosov-2026 International Conference, Astana; 2026

- Built a regression model of residential property prices in Astana from real-estate listings using ArcGIS spatial analysis; identified location, infrastructure proximity and environmental factors as key price drivers
- 3rd place of 20 reports, Geo-ecology and Spatial Analysis section

Nazarbayev University Summer Research Program; 2025

- Analysed datasets to test venture hypotheses, wrote the final report and pitched it to a panel; audited BUS101

EXPERIENCE

PrimeTeens, Curriculum Designer, Mentor & Platform Developer; Jun 2026 – present

- Built the platform end to end: Next.js/TypeScript site and live student workbook (SSE), Python Telegram bots (aiogram, Whisper voice input), PostgreSQL/Prisma, deployed on Linux with nginx, systemd, backups and monitoring
- Designed and taught a research-and-portfolio curriculum for grades 7–11 (research methods, debate and Model UN, hackathon finale) to 6 groups; produced the full teaching kit: decks, workbook, 6 hackathon cases, 16 guides

Coventry University Kazakhstan, Library Automation Intern; Jan 2026 – present

- Built a QR-based book issuance system (low-cost alternative to RFID), the library website and a RAG chatbot for it

"Anisa Spectrum" Limited, Software Developer; Sep 2024 – Dec 2024

- Developed an app for early autism diagnosis, now deployed in a local hospital: UI/UX design, PostgreSQL database

InUniGlobal, Digital Marketing Intern; Apr 2024 – May 2024

- Redesigned the company's Instagram profile (still in use); published daily posts and stories designed in Figma & PS

UNICEF Kazakhstan, Volunteer Computer Science Teacher; Jan 2024 – May 2024

- Remotely mentored children from disadvantaged backgrounds through their first programming project

PROJECTS

- **Shiftray** (open source): ASUS laptop control app for Linux; Python/Qt, Fedora COPR, Arch, AppImage, 7 languages
- **Bookgram** (in development): social reading app with a React Native/Expo client and a Fastify/TypeScript API on PostgreSQL (Prisma), Redis and Socket.IO
- **bloody-w95-linux**: reverse-engineered a gaming mouse's USB protocol; RGB and DPI control from Linux

VOLUNTEERING & LEADERSHIP

EcoHackathon & Bright Future Hackathon, Main Organizer; Jan 2024 – Aug 2024

- Ran two national hackathons across 14 cities (650+ participants, 175 teams): logistics, rules, tasks and marketing

WeCare Volunteer Club, Co-leader; 2022 – 2024

- Coordinated 50+ volunteers for weekly events at 3 animal shelters

HONORS & AWARDS

1st place, BICAP Hackathon 2024 (200 teams) · 3rd place, National Economic Challenge 2024 (125 teams) · 3rd place, InsurTechHackathon 2023 (150 teams) · Top 20 of 400 teams, AmanatTech · State Scholarship of Kazakhstan

SKILLS

Programming: Python, TypeScript, SQL, HTML/CSS; basic R and MATLAB
Frameworks: FastAPI, SQLAlchemy, Next.js, React Native, Prisma, aiogram, PySide6 (Qt), Playwright
Tools & data: PostgreSQL, Redis, InfluxDB, Docker, Git, Linux, NumPy, Pandas, Matplotlib, ArcGIS; ML basics
Languages: Russian (native), English (fluent), Kazakh (intermediate), German (A1)